Estimation of Change in Population of Municipalities in PR

[ESMA 6835]: Bayesian Inference

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Contents

1 Introduction

Changes in demographics are a highly studied topic and are of great interest to governments. Understanding demographic characteristics is particularly important for regional governments, such as the municipalities in Puerto Rico. Anticipating how the population will change in the coming years is fundamental for planning infrastructure projects and public investments, especially in areas experiencing population growth, decline, or aging populations.

Classical inference methods, such as classical linear regression, are frequently used to study these trends. However, applying Bayesian inference methods offers significant advantages. This is due to the ability to incorporate prior information and provide complete probability distributions for the estimated parameters.

Based on these arguments, we are led to ask the following questions: How has the population of a municipality in Puerto Rico changed over time? Therefore, through this project, we will use simple linear regression with a Bayesian approach to estimate and analyze population change in Puerto Rican municipalities over time.

2 Methodology

2.1 Data

This research uses data from the American Community Survey 5-year estimates (ACS) produced by the U.S. Census Bureau. The dataset includes population estimates from 2011 to 2023 for all 78 municipalities in Puerto Rico.

The variables to consider for our simple regression model are: the years for which we have population estimates for Puerto Rico (independent variable X) and the total population for the 78 municipalities of Puerto Rico for each year (dependent variable Y).

2.2 Regression Model

The simple linear regression model to be used is the following:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

where:

- Y_i = population in year i
- X_i = year
- β_0 = intercept
- β_1 = population change rate
- ϵ_i = random error

2.3 Bayesian Approach

Due to the nature of the Bayesian approach, we will use non-informative prior distributions. However, we will use the Markov Chain Monte Carlo method to obtain the posterior distributions of the data. This process will be executed in Python software.

3 Results

A classical linear regression was performed with a significance level of $\alpha = 0.05$, from which we obtained a population change rate of -210.52, indicating that the total population decreases each year. However, upon reviewing the p-value of this test (0.540), we realized that our findings are not statistically significant. Therefore, based on classical inference, there is insufficient evidence to determine that there is a drastic change in the population decrease over time in this model. Moreover, since the adjusted R-squared is -0.001 this indicates that the model with the variable time might not be the best to explain the changes in populations.

Regarding the Bayesian regression, we obtained that the mean of the posterior distribution of the coefficient corresponding to the years is -211.04. This indicates that the population decreases per year. Furthermore, reviewing the Bayesian confidence intervals, it was found that there is a 94% probability that the year effect is between -822.54 and 444.45. According to the credibility intervals, with 94% confidence, the expected coefficient for years is between -500.00 and 50.00. Because this interval includes zero, there is no statistically significative probability that the effect of years is negative. Regarding the MCMC chains, the chains appear to blend quite well, and the R-squared value is approximately 1, suggesting that the model converged correctly. Although both classical and Bayesian inference essentially conclude that the population rate of change and the model do not answer our question, the Bayesian inference method is more robust. This is due to the fact that the Bayesian method uses prior information to perform the analyses. This allows the model to be normalized, reduces model identification problems, and makes the estimators less unstable.

4 Predicciones

Due to the lack of statistical significance in the results regarding changes over time in municipal population, no meaningful recommendations can be made regarding the problem.

5 Conclusions

The overall results of the simple regression are inconclusive, as the variable representing year was not statistically significant. This lack of statistical significance may be due to the limitations of simple regression models. A major limitation is the violation of the assumption of independence and identical distribution (i.i.d.), since population values are highly correlated with those of the previous year. It is advisable to use a more appropriate model for time series data, such as an ARIMA model.

Appendix A: Simple regression

Table 1: OLS Regression Results

Variable	**Coefficient**	**Std. Error**	**t-Statistic**	**P-value**	**95% CI**
Intercept	461,000	693,000	0.666	0.506	[-899,000, 1.82M]
Year	-210.515	343.336	-0.613	0.540	[-884.247, 463.216]
Model Fit					
R-squared	0.000				
Adjusted R-squared	-0.001				
F-statistic	0.3759			0.540	
Prob (F-statistic)				0.540	
No. Observations	1014				
Diagnostics					
Omnibus	898.513			0.000	
Jarque-Bera	22,245.704			0.000	
Skew	4.144				
Kurtosis	24.397				
Durbin-Watson	1.935				
Condition No.	1.09e+06				

Table 2: Bayes simple regression Results (Mean, SD, HDI)

Variable	**Mean**	**SD**	**HDI 3%**	**HDI 97%**
Intercept	479,133.653	704,469.055	-922,695.260	1,743,484.519
Sigma	40,914.524	917.309	39,261.811	42,711.419
Year	-219.481	349.280	-847.946	473.013

Table 3: Bayes simple regression Results (MCSE, ESS, and R-hat)

Variable	MCSE Mean	MCSE SD	ESS Bulk	ESS Tail	R-hat
Intercept	9,270.616	11,465.597	5,768.0	2,821.0	1.0
Sigma	11.705	14.697	6,165.0	3,085.0	1.0
Year	4.597	5.685	5,767.0	2,821.0	1.0

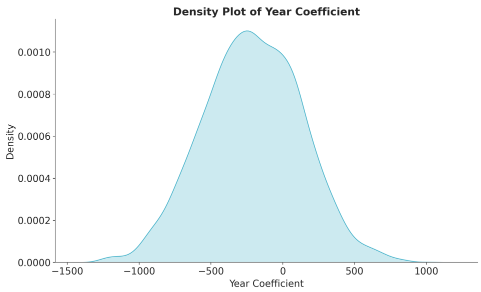

Figure 1: Density of simple regression

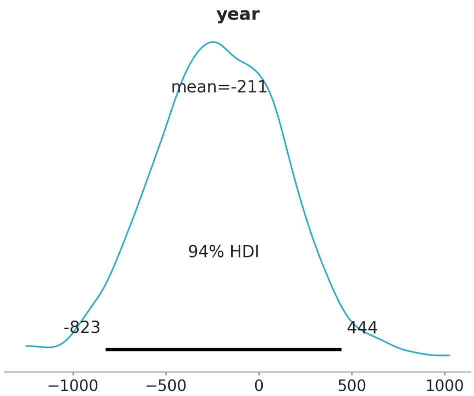

Figure 2: Confidence interval of Simple regression

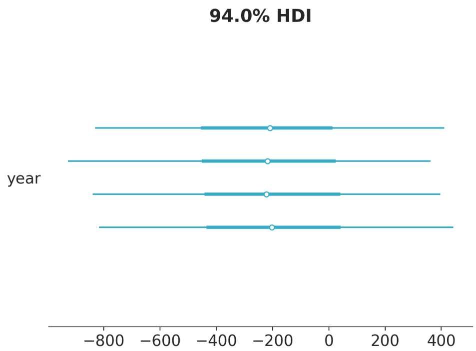

Figure 3: Credibility interval of Simple regression

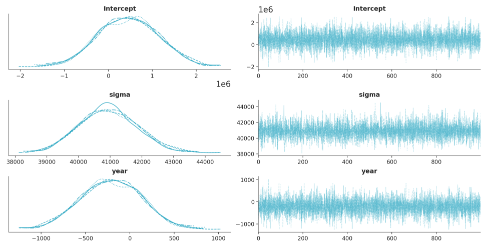

Figure 4: Trace graph of simple regression

Appendix B: Log regression

Table 4: OLS Regression Results (Model Fit and Coefficients)

Variable	Coef	Std. Error	t-Statistic	P-value	95% CI
Intercept	20.8794	13.317	1.568	0.117	[-5.254, 47.012]
Year	-0.0053	0.007	-0.804	0.422	[-0.018, 0.008]
Model Fit					
R-squared	0.001				
Adjusted R-squared	-0.000				
F-statistic	0.6461			0.422	
Prob (F-statistic)				0.422	
No. Observations	1014				

Table 5: Bayesian Sampling Results (Mean, SD, HDI)

Variable	Mean	SD	HDI 3%	HDI 97%
Intercept	20.846	13.403	-4.771	46.100
Sigma	0.788	0.018	0.756	0.822
Year	-0.005	0.007	-0.018	0.007

Table 6: Bayesian Sampling Results (MCSE, ESS, and R-hat)

Variable	MCSE Mean	MCSE SD	ESS Bulk	ESS Tail	R-hat
Intercept	0.17	0.226	6221.0	2891.0	1.0
Sigma	0.00	0.000	6424.0	2900.0	1.0
Year	0.00	0.000	6217.0	2865.0	1.0

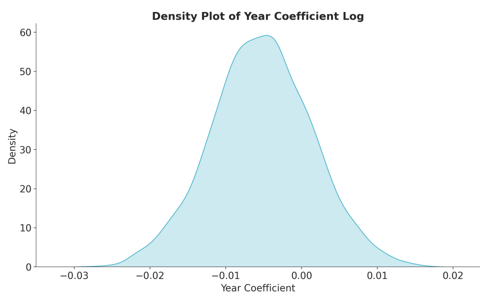

Figure 5: Density of Log regression

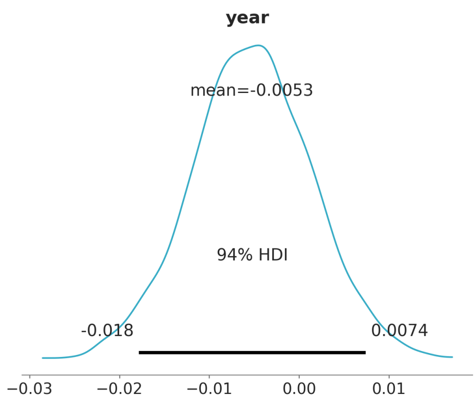

Figure 6: Confidence interval of Log regression

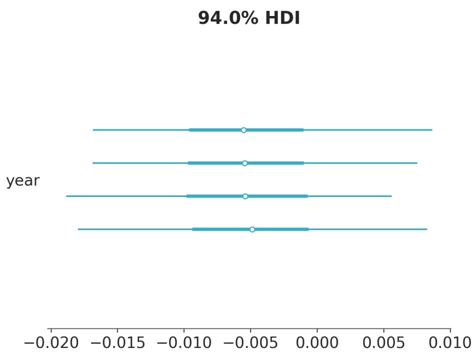

Figure 7: Credibility interval of Log regression

6 Appendix C: Log fixed effect

Table 7: OLS Regression Results (Model Fit and Initial Coefficients)

Variable	**Coef**	**Std. Error**	**t-Statistic**	**P-value**	**95% CI**
Intercept	20.3180	0.517	39.291	0.000	[19.303, 21.333]
C(county)[T.3.0]	0.7826	0.012	65.330	0.000	[0.759, 0.806]
C(county)[T.5.0]	1.1297	0.012	94.300	0.000	[1.106, 1.153]
C(county)[T.7.0]	0.3701	0.012	30.898	0.000	[0.347, 0.394]
C(county)[T.9.0]	0.2939	0.012	24.537	0.000	[0.270, 0.317]
C(county)[T.11.0]	0.4228	0.012	35.290	0.000	[0.399, 0.446]
Model Fit					
R-squared	0.999				
Diagnostics					
AIC	-4122				
BIC	-3733				
Df Residuals	935				
Df Model	78				
Condition No.	1.09e+06				

Table 8: Bayesian Fixed Effect Results (Mean, SD, HDI)

Variable	Mean	SD	HDI 3%	HDI 97%
Intercept	20.444	13.228	-3.724	45.340
C(county)	0.000	0.001	-0.001	0.001
Sigma	0.788	0.017	0.755	0.821
Year	-0.005	0.007	-0.017	0.007

Table 9: Bayesian Fixed Effect Results (MCSE, ESS, and R-hat)

Variable	MCSE Mean	MCSE SD	ESS Bulk	ESS Tail	R-hat
Intercept	0.155	0.201	7236.0	3122.0	1.0
C(county)	0.000	0.000	6710.0	3373.0	1.0
Sigma	0.000	0.000	7065.0	3154.0	1.0
Year	0.000	0.000	7238.0	3049.0	1.0

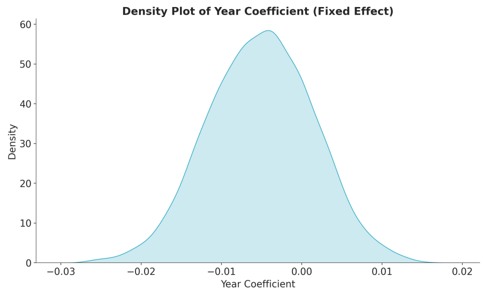

Figure 8: Density of fix effect regression

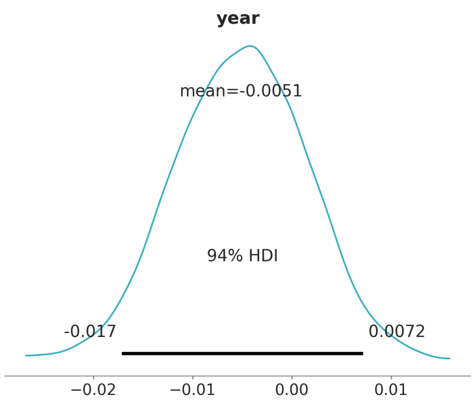

Figure 9: Confidence interval of fix effect regression

Table 10: Bayesian Sampling Results (MCSE, ESS, and R-hat)

Variable	**MCSE Mean**	**MCSE SD**	**ESS Bulk**	**ESS Tail**	**R-hat**
Intercept	0.17	0.226	6221.0	2891.0	1.0
Sigma	0.00	0.000	6424.0	2900.0	1.0
Year	0.00	0.000	6217.0	2865.0	1.0

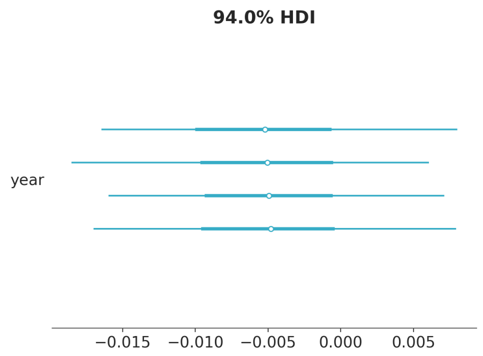

Figure 10: Credibility interval of Fix effect regression